GCP-GraphRAG Research Report

Generated: 2026-05-12 15:01
Query: What are the drone regulations in India in 2023?
Embedding model: sentence-transformers
Generation model: phi3:mini

1. Pipeline Configuration

Parameter	Value
Relevance weight	0.5
Coherence weight	0.3
Provenance weight	0.2
Top-K retrieval	8
FAISS index	flat
Edge threshold	0.4
GAT layers	2
Attention heads	4
Context window	4096 tokens
GCP solver	greedy

2. Evaluation Metrics

Metric	Score
F1 Score	100.0%
ROUGE-L	100.0%
Recall@5	60.0%
Evidence Density	62.5%
Budget Adherence	2.2%
Context Reduction	0.0%
Redundancy Ratio	0.0%
Relevance / Token	0.0504

3. Comparison with Existing Research

System	F1	ROUGE-L	Recall@5	Context Reduction
Naive RAG (BM25 + T5)	42.1%	38.9%	51.2%	0.0%
Dense RAG (DPR + GPT2)	53.8%	49.7%	63.4%	0.0%
GraphRAG (baseline)	61.2%	57.3%	70.1%	18.0%
mDeBERTa + mBART (paper prop.)	68.1%	63.4%	75.2%	31.0%
GCP-GraphRAG (this run)	**100.0%**	**100.0%**	**60.0%**	**0.0%**

> *Baseline figures sourced from published RAG literature. Bold row = current system result.*

4. Generated Response

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In 2023, India's drone regulations were updated to require remote ID for drones over 25
  0g. Additionally, geo-fencing and real-time tracking became mandatory for commercial
  operators. Agricultural drones may operate in Green Zones without prior permission. A
  Remote Pilot Certificate is required for drones weighing more than 2 kg. Drones are c
  ategorized by weight: micro (≤250g), small (250g–2kg), medium (2–25kg), and large (>2
  5kg). No-fly zones include areas within 5 km of airports, 3 km of defence establishme
  nts.
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5. Evidence Summary

Total chunks retrieved: 8 / 8
Tokens used: 89 / 3987
Latency: 124423.1 ms

#	Source	Modality	Relevance
1	dgca*gov*in	text	1.000
2	meity_2023	text	0.909
3	hindi_gazette	text	0.984
4	meity_2023	text	0.396
5	agriculture_ministry	text	0.599
6	aviation_ministry	text	0.536
7	dgca*gov*in	text	0.000
8	pib*gov*in	text	0.063

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6. Graph Statistics

Metric	Value
Nodes	8
Edges	2
Density	0.071

Report generated by GCP-GraphRAG Research Platform